

II SEM DS LONG ANSWER QUESTIONS

Unit – I

1. Explain Classification of Data Structures & Explain Various operations on Data Structures.
2. Explain Merge Sort algorithm with an example.
3. Explain Linear Search algorithm with an example.
4. Explain about various Sorting methods.
5. Explain Quick Sort algorithm with an example.

Unit – II

1. What is Linked List? Explain various Operations of Linked List.
2. Write about insertion and deletion operations of double linked List.
3. Write about Circular Linked List.
4. What are the applications of Linked List? Explain about polynomial addition using linked list.

Unit – III

1. Define Stack. Explain various operations performed on Stack.
2. Explain the operations of Stack using Array.
3. Explain the implementation of Stack using Linked List.
4. Explain algorithm for Infix to Post fix conversion with an example.
5. Explain applications of Stack. Write an algorithm for reversing a List.

Unit – IV

1. Define Queue. Explain various operations of Queue.
2. Explain the implementation of Queue using Array.
3. Explain the implementation of Queue using Linked List.
4. Explain various operations on deque(double ended queue).
5. Explain various applications of Queue. Explain BFS algorithm with an example.

Unit – V

1. Define Binary Tree. Explain various Binary Tree Representations (Array, Linked List) with examples.
2. Explain various Binary Tree Traversal algorithms with examples.
3. Define BST. Explain the construction of BST with an example.
4. Explain procedure to delete a node(with 0,1,2 Childs) in BST.
5. Explain various Hashing Functions.
6. Explain various collision resolution methods.
7. Explain Separate chain & Linear probing methods with examples.